

Basic Statistics

BCA — Semester 2 — — Answer Sheet

Subject Code:

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

1. Distinguish between descriptive and inferential statistics. The following table represents the mark obtained by a batch of 10 students in a certain class tests in Basic Statistics and Computer science. Marks in Basic Statistics 53 55 52 32 30 60 47 46 35 58 Marks in Computer science 57 45 24 31 25 84 43 80 32 72 Indicate in which subject is the level of more consistency?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Define regression. From the following table, compute the line of regression for estimating blood pressure: Blood pressure Y 147 125 160 118 149 128 Age in years X 56 42 72 36 63 47 (i) Fit a regression equation that best describe the above data. (ii) Estimate the blood pressure when age is 50 yrs. (iii) Interpret the regression coefficient.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Under which situation Normal distribution will be used. The annual salaries of employees in a large company are approximately normally distributed with a mean of \$50,000 and a standard deviation of \$20,000. (a) What is the probability of people earns less than \$40,000? (b) What is the probability of people earns between \$45,000 and \$65,000? (c) What is the probability of people earns more than \$70,000?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. Describe the role of Statistics in information technology.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. The following table shows the mode of transport used by 400 students of a school. Represent the following information on the bar diagram and Pareto diagram. Mode of transport Bus Bicycle On foot By car No. of students 200 100 80 20

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Write properties of correlation. A study is made relating aptitude scores to productivity in a factory after six months training of personnel. The following are the figures regarding six randomly selected workers:
Aptitude Scores X: 9 18 18 20 20 23
Productivity Index Y: 12 33 23 42 29 30
Find the coefficients of correlation between aptitude score and productivity index and comment on the value. Also find coefficient of variation for each variable.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Define random variable. A random variable X has following probability functions.
 $P(X=0)=0.1$, $P(X=1)=0.22$, $P(X=2)=0.3$
Find, (i) The value of C. (ii) $P(X < 3)$, $P(X \geq 1)$. (iii) Mean and variance.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Define binomial distribution. Fit a binomial distribution on the following data:
x: 0 1 2 3 4
f: 28 62 46 10 4

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. The probability that an integrated circuit chip will have defective etching is 0.12, the probability that it will have a crack defect is 0.29, and the probability that it has both defects is 0.07. What is the probability that a newly manufactured chip will have either an etching or crack defect.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. If the first four moments about mean are 0, 2.8, -2 and 24.5 respectively. Compute coefficient of skewness and kurtosis and comment upon result.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. The fuel consumption of a new model of cars is being tested. In one trial, 50 cars chosen at random were driven under the identical conditions and the distances, x km. covered on 1 liter of petrol were recorded. The results gave the following totals: $\sum x = 525$, $\sum x^2 = 5625$. Calculate the 99% confidence interval for the mean petrol consumption, in km per liter. Interpret the result.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Write short note on the following:
a) Use of Box and whisker plot.
b) Parameter and Statistic.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.